Adaptive Extended Kalman Filter for State-of-Charge Estimation Across Battery Aging States at the Cell Level

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***Abstract*—This paper presents a complete pipeline for State of Charge (SOC) estimation of lithium ion cells across five aging states spanning 100%–80% State-of-Health (SOH). The Doyle–Fuller–Newman (DFN) electrochemical model with SEI growth, lithium plating, and loss of active material (Marquis2019 [1], NMC/Graphite, 0.68 Ah) generates aging data over 100 cycles. Five SOH snapshots are characterised via quasi-static OCV curves and Hybrid Pulse Power Characterisation (HPPC), and a 2RC Equivalent Circuit Model (ECM) is fitted as SOH×SOC lookup tables. An Adaptive Extended Kalman Filter (AEKF) with Sage Husa noise adaptation [5] and voltage-bias compensation is validated on EPA UDDS and HWFET drive cycles. All ten test cases achieve SOC RMSE below 2.5% with the baseline AEKF and below 1% following PSO-based noise optimisation, with convergence under one second from a 15-percentage-point initial error.**

***Index Terms*—State-of-Charge estimation, Adaptive Extended Kalman Filter, Sage–Husa adaptation, battery aging, Doyle–Fuller–Newman model, equivalent circuit model, HPPC parameter identification**

I. Introduction

A. Background and Motivation

Lithium-ion batteries are the dominant energy storage technology for electric vehicles (EVs), stationary storage, and portable electronics. As cells age, two principal mechanisms degrade performance: capacity fade (loss of cyclable lithium) and power fade (increased internal resistance). Traditional Battery Management System (BMS) SOC estimators are parameterised at beginning-of-life (BOL). As aging progresses, internal resistance rises 50 to 100% and capacity falls to ∼80% SOH at end-of-life (EOL). This parameter mismatch causes the filter to generate incorrect voltage predictions, which it misinterprets as SOC error, producing systematic drift often exceeding 10% by EOL. This directly threatens range prediction reliability and BMS safety margins.

B. Objectives

This work addresses that gap with a five-task pipeline:
(i) DFN aging simulation producing electrochemically consistent SOH snapshots.
(ii) SOH-resolved OCV and HPPC characterisation.
(iii) 2RC ECM parameter identification as lookup tables.
(iv) Ground-truth drive cycle generation.
(v) AEKF implementation and validation across the full BOL–EOL range.

C. Report Organisation

Section II covers the battery modeling pipeline. Section III presents the AEKF estimator design and tuning. Section IV presents simulation results and discussion. Section V concludes with a summary and future directions. Section V-B outlines team contributions.

II. Battery Modeling

A. System Description

The cell under study is an NMC/Graphite chemistry with nominal capacity 0.68 Ah and operating voltage window 3.2–4.2 V. All electrochemical parameters were taken from the Marquis2019 dataset [1]. Simulations were executed in PyBaMM 25.12.2 [2] using Python 3.12.10.

The graphite anode hosts lithium intercalation: $\mathrm{Li}^+ + e^- + \mathrm{C}_6 \rightleftharpoons \mathrm{LiC}_6$, and the NMC cathode: $\mathrm{LiNiMnCoO}_2 \rightleftharpoons \mathrm{Li}_{1-x}\mathrm{NiMnCoO}_2 + x\mathrm{Li}^+ + xe^-$.

B. Mathematical Formulation — DFN Model

The Doyle–Fuller–Newman (DFN) model [3] resolves electrode-level physics through four coupled PDEs.

1) Solid-Phase Diffusion: Lithium concentration c_s in each spherical electrode particle:

$$\frac{\partial c_s}{\partial t} = \frac{D_s}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial c_s}{\partial r}\right), \quad -D_s\left.\frac{\partial c_s}{\partial r}\right|_{r=R_p} = \frac{j_{\mathrm{main}}}{F} \tag{1}$$

2) Electrolyte Diffusion:

$$\frac{\partial(\varepsilon_e c_e)}{\partial t} = \frac{\partial}{\partial x}\left(D_e^{\mathrm{eff}}\frac{\partial c_e}{\partial x}\right) + a(1-t_+)\,j \tag{2}$$

3) Butler–Volmer Kinetics:

$$j_{\mathrm{main}} = i_0\left[\exp\left(\frac{\alpha_a F\eta}{RT}\right) - \exp\left(-\frac{\alpha_c F\eta}{RT}\right)\right] \tag{3}$$

4) Terminal Voltage:

$$V_t = \phi_s^+ - \phi_s^- - \eta_{\mathrm{losses}} \tag{4}$$

C. SEI Growth and Degradation Mechanism

At the graphite anode ($\sim$0.1 V vs. Li/Li$^+$), ethylene carbonate (EC) undergoes parasitic reduction, depositing Li_2CO_3 as the SEI layer:

$$2\,\mathrm{EC} + 2\,\mathrm{Li}^+ + 2e^- \rightarrow \mathrm{Li_2CO_3} + \mathrm{CH_2{=}CH_2} \uparrow \tag{5}$$

EC must diffuse through the growing film to sustain the reaction. The SEI reaction current with EC-limited diffusion is:

$$j_{\mathrm{SEI}} = -\frac{i_{0,\mathrm{SEI}}}{RT} \exp\left(-\frac{\alpha_c F \eta_{\mathrm{SEI}}}{RT}\right) \cdot \frac{D_{\mathrm{EC}}\, c_{\mathrm{EC,bulk}}}{\delta_{\mathrm{SEI}}} \tag{6}$$

Since the rate is inversely proportional to the film thickness, this produces the self-limiting $\sqrt{t}$ growth law: $\delta_{\mathrm{SEI}} \propto \sqrt{t}$, yielding the characteristic square-root capacity fade in real cells. SOH is tracked as:

$$\mathrm{SOH}(N) = \frac{Q_0 - F\int_0^N \int_\Omega |j_{\mathrm{SEI}}|\, a\, d\Omega\, dt}{Q_0} \tag{7}$$

Three mechanisms were enabled: SEI growth (early cycles), lithium plating (high C-rates), and loss of active material (LAM, particle cracking). Using all three ensures a physically realistic degradation trajectory across 100 cycles.

Open-source model modification: The negative electrode active material volume fraction $\varepsilon_s^{\mathrm{neg}}$ was scaled proportionally to SOH for the OCV characterisation task to represent lithium inventory loss: $\varepsilon_s^{\mathrm{neg}}\big|_{\mathrm{SOH}} = \varepsilon_s^{\mathrm{neg}}\big|_{\mathrm{BOL}} \times \mathrm{SOH}/100$. All other Marquis2019 parameters are unmodified.

D. Simulation Setup and SOH Milestones

Each of 100 cycles used 1C CC charge to 4.2 V, CV hold to C/50, and 1C CC discharge to 3.2 V. Cycle 1 included an additional formation charge and CV hold prior to the first discharge to condition the cell. At each SOH milestone the full PyBaMM internal electrochemical state was extracted as a `starting_solution` and used to initialise the HPPC tests and drive cycle simulations at the correct aging state.

Fig. 1 shows the full SOH degradation trajectory, confirming the characteristic concave fade profile predicted by the $\sqrt{t}$ SEI growth law.

TABLE I: SOH Milestones from 100-Cycle DFN Simulation

SOH (%)	Cycle	Capacity (Ah)	State
100	1	0.6806	BOL
95	47	0.6466	Mild aging
90	65	0.6125	Moderate
85	81	0.5785	Aged
80	95	0.5445	EOL

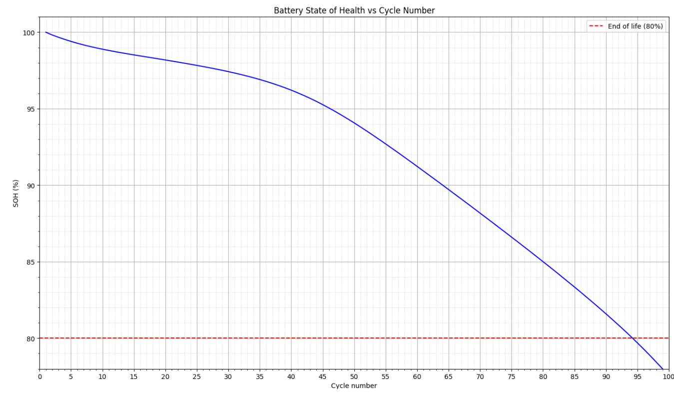

Fig. 1: SOH degradation trajectory over 100 charge–discharge cycles. The concave shape reflects the $\sqrt{t}$ self-limiting SEI growth law rapid early capacity loss that progressively slows as the SEI film thickens. The red dashed line marks the 80% EOL threshold, reached at cycle 95.

E. OCV–SOC Characterisation

Five OCV–SOC curves were generated by running a fresh C/25 quasi-static characterisation at each SOH state. Aging was represented by scaling the negative electrode active material volume fraction proportionally to SOH ($\varepsilon_s^{\mathrm{neg}}\big|_{\mathrm{SOH}} = \varepsilon_s^{\mathrm{neg}}\big|_{\mathrm{BOL}} \times \mathrm{SOH}/100$), capturing lithium inventory loss (LLI) without requiring the aged `starting_solution`.

The characterisation protocol comprised a full 1C charge to 4.2 V, CV hold to C/50, C/25 discharge to 3.2 V, 3600 s rest, C/25 charge to 4.2 V, and a final 3600 s rest. The true OCV was computed as the pointwise average of the discharge and charge voltage curves, cancelling the small hysteresis present at C/25.

As seen in Fig. 2, the curves diverge most at high SOC (near 1.0) and low SOC (near 0.0), where lithium inventory loss has the greatest effect on electrode stoichiometry. The mid-SOC region (0.4–0.6) shows less separation because both electrodes still operate within their usable stoichiometric windows. In the AEKF, OCV is retrieved at runtime via a `scipy.interpolate.RegularGridInterpolator` over a 2D (SOH, SOC) grid, enabling smooth interpolation between the five measured curves.

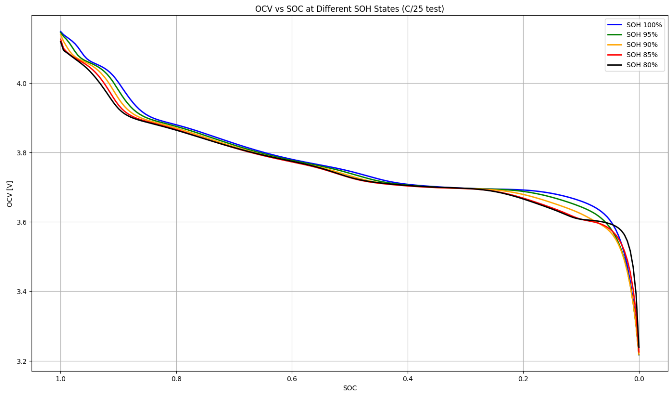

Fig. 2: OCV–SOC curves at all five SOH states from C/25 quasi-static characterisation (discharge–charge average). Curves shift leftward with aging at any given voltage, the aged cell appears more depleted. Divergence is most pronounced at high and low SOC where lithium inventory loss dominates electrode stoichiometry.

F. HPPC Testing

HPPC tests were simulated at each of the five SOH states, initialised from the corresponding aged `starting_solution`. The protocol repeated the following pulse sequence at nine SOC levels stepping from 90% down to 10%:

1) 600 s rest full electrochemical relaxation to steady state.
2) Discharge pulses: 0.5C, 1C, 2C each for 30 s, separated by 600 s rests.
3) Charge pulses: 0.5C, 1C each for 30 s, separated by 600 s rests.
4) 1C discharge for 360 s steps SOC down to the next level.

The 600 s rest periods ensure the cell reaches equilibrium before each pulse, providing clean relaxation transients for RC parameter fitting. Positive current denotes discharge throughout (PyBaMM sign convention). The 2C charge pulse was excluded as it caused voltage cutoff violations at lower SOH states. Fig. 3–7 show the HPPC voltage and current traces across all five SOH states. The nine SOC levels are visible as the stepped decline in resting voltage from left to right. Voltage excursions grow progressively larger at lower SOC levels, reflecting increased electrode polarisation as lithium inventory depletes. As SOH decreases from 100% to 80%, the voltage window narrows and pulse excursions under identical current inputs grow larger, consistent with the monotonic increase in R_0 with aging.

Fig. 3: HPPC test at SOH 100% (BOL). Voltage (blue, left axis) and current (red, right axis) vs. time.

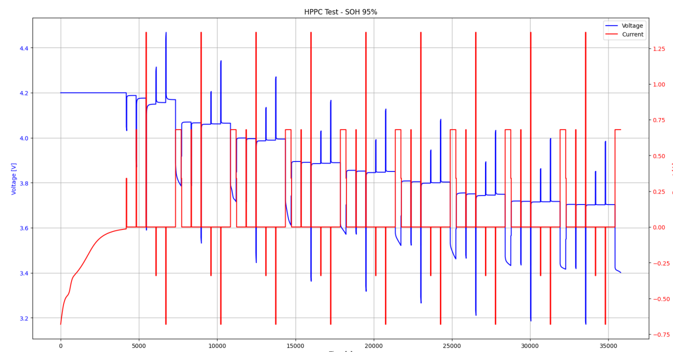

Fig. 4: HPPC test at SOH 95%.

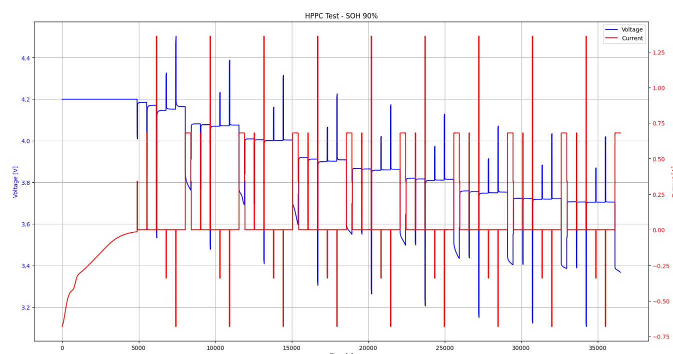

Fig. 5: HPPC test at SOH 90%.

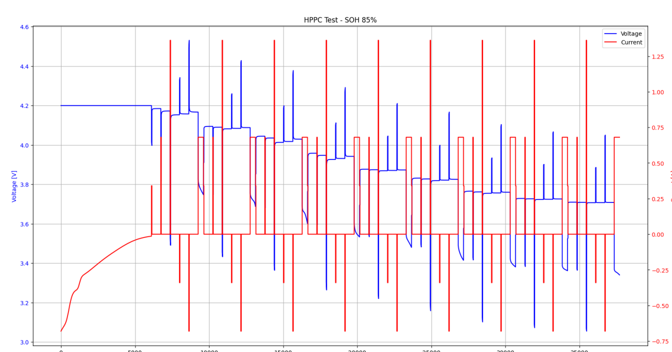

Fig. 6: HPPC test at SOH 85%.

Fig. 7: HPPC test at SOH 80% (EOL). Narrowest voltage window and largest pulse excursions across all five states, consistent with maximum R_0 at end of life.

G. ECM Parameter Identification

1) Parameter Extraction: R_0 was extracted from the instantaneous voltage jump at the onset of each 1C discharge pulse:

$$R_0 = \left.\frac{\Delta V}{\Delta I}\right|_{t=0^+} \tag{8}$$

RC parameters were extracted by fitting a double-exponential model to the post-1C-discharge rest segment:

$$V_{\text{relax}}(t) = V_\infty - A_1 e^{-t/\tau_1} - A_2 e^{-t/\tau_2} \tag{9}$$

with constraints $\tau_1 \in [1, 50]$ s (fast RC branch) and $\tau_2 \in [50, 600]$ s (slow RC branch), giving $R_i = A_i/\Delta I$ and $C_i = \tau_i/R_i$. All parameters are stored as 5×9 SOH×SOC lookup tables.

2) R0 Results: Figs. 8–12 show R_0 vs. SOC at each SOH state. R_0 exhibits a characteristic U-shape with SOC at every aging level elevated at high SOC, reaching a minimum near SOC 0.6–0.7, then rising sharply at low SOC as electrode kinetics slow near full depletion. Fig. 13 confirms that average R_0 increases monotonically from 0.168 Ω at SOH 100% to 0.521 Ω at SOH 80% a 3.1× increase driven by SEI film thickening.

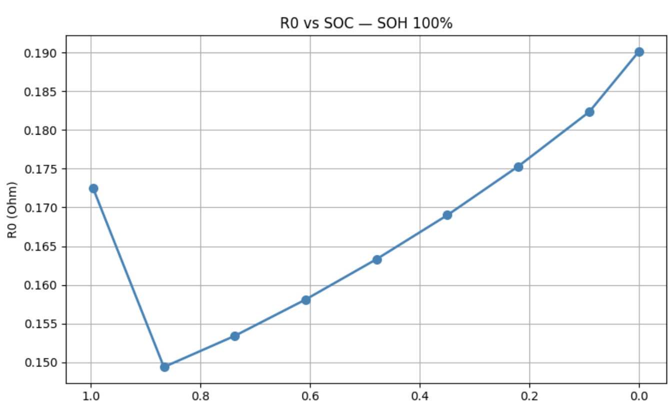

Fig. 8: R_0 vs. SOC at SOH 100% (BOL). Average $R_0 = 0.168$ Ω.

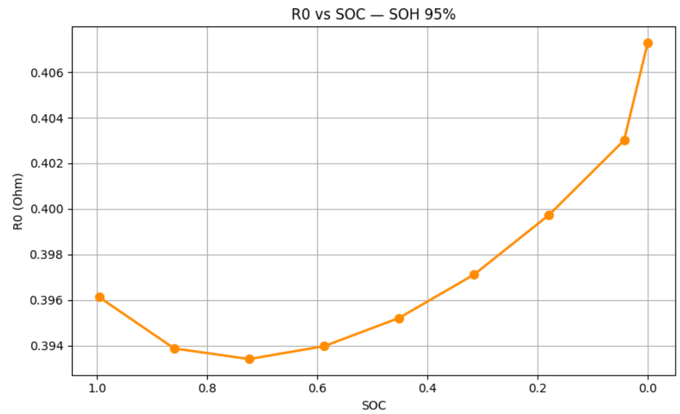

Fig. 9: R_0 vs. SOC at SOH 95%. Average $R_0 = 0.398$ Ω.

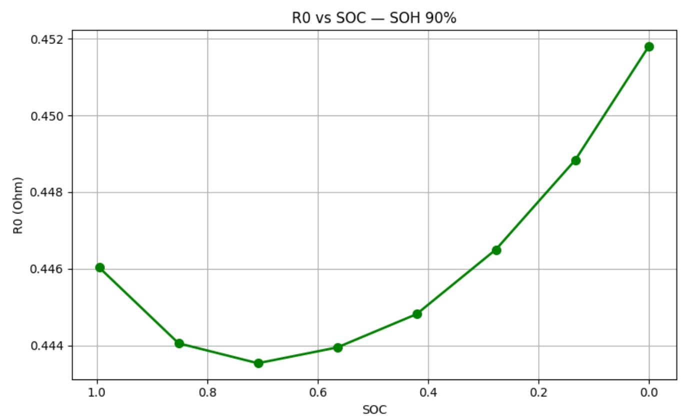

Fig. 10: R_0 vs. SOC at SOH 90%. Average $R_0 = 0.447$ Ω.

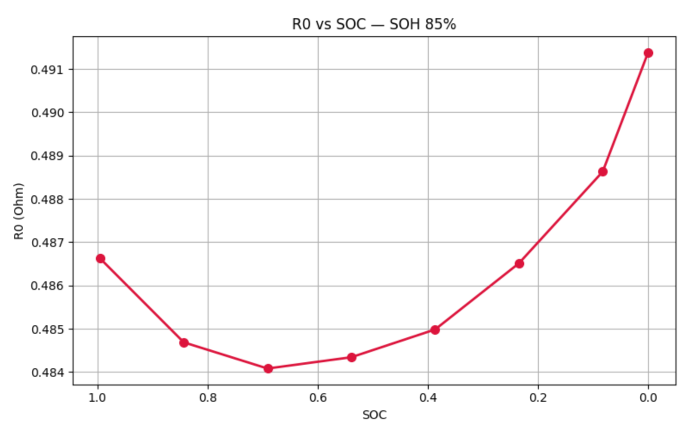

Fig. 11: R_0 vs. SOC at SOH 85%. Average $R_0 = 0.487$ Ω.

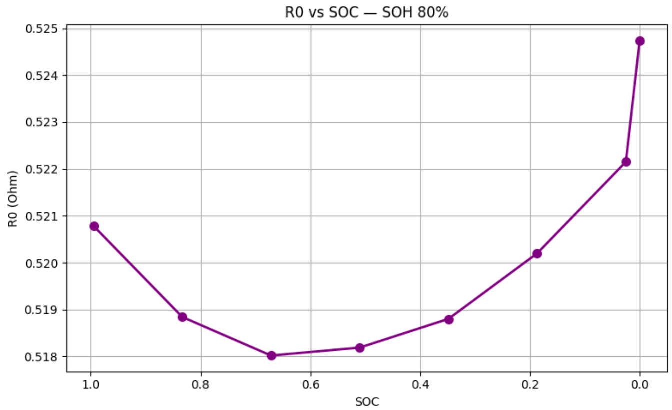

Fig. 12: R_0 vs. SOC at SOH 80% (EOL). Average R_0 = $0.521\ \Omega$. The U-shape is consistent across all aging states.

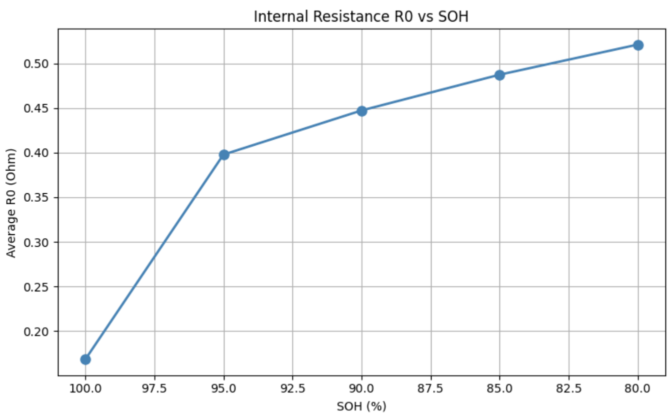

Fig. 13: Average R_0 vs. SOH. Monotonically increasing from BOL to EOL, confirming progressive SEI thickening as the dominant ohmic degradation mechanism.

3) RC Parameter Results: Figs. 14–18 show the RC parameters. R_1 and R_2 are non-monotonic with both SOC and SOH a fundamental artifact of projecting distributed porous-electrode PDE dynamics onto a lumped 2RC structure [6]. As aging creates spatial non-uniformity in electrode microstructure, the relative contribution of fast and slow relaxation modes shifts in ways a lumped RC pair cannot track monotonically. C_1 and C_2 show a general decreasing trend with aging when averaged across SOC (Fig. 18), consistent with reduced effective double-layer capacitance, but exhibit local scatter at individual SOC points due to the same structural mismatch. The AEKF's Sage–Husa adaptive noise tuning handles this limitation at runtime rather than forcing artificial monotonicity.

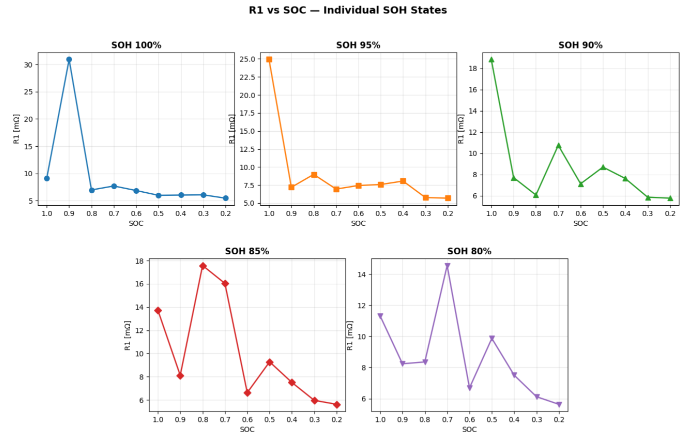

Fig. 14: R_1 vs. SOC across all five SOH states. Non-monotonic variation with both SOC and SOH reflects the DFN→ECM structural mismatch.

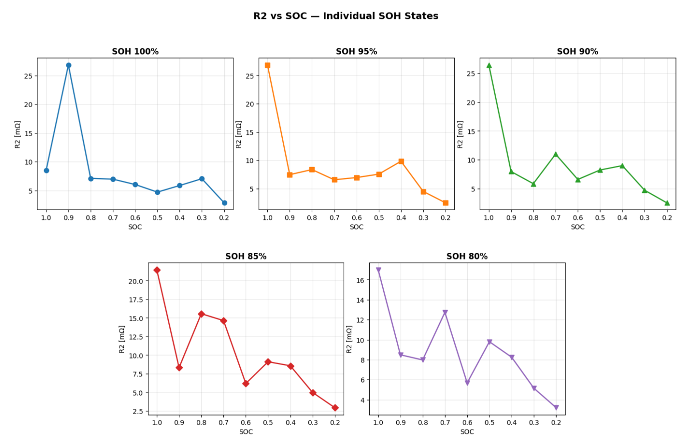

Fig. 15: R_2 vs. SOC across all five SOH states. Similar non-monotonic behaviour to R_1.

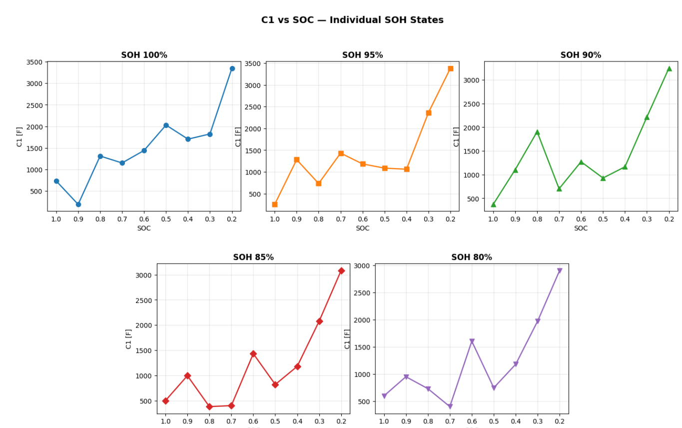

Fig. 16: C_1 vs. SOC across all five SOH states. Values generally increase toward low SOC (longer time constants at low lithium content) but show scatter across SOH states.

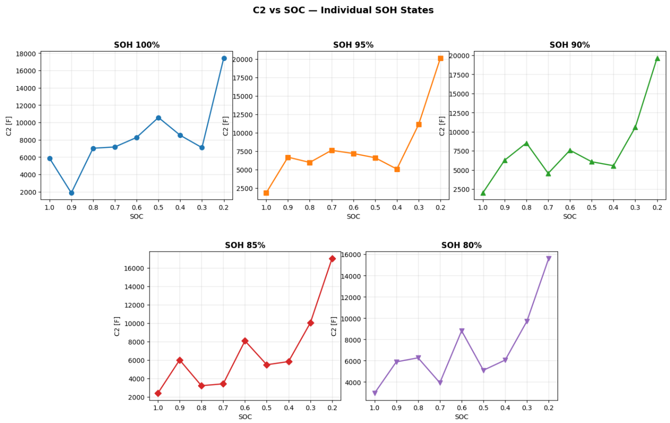

Fig. 17: C_2 vs. SOC across all five SOH states. Slow branch capacitance follows a similar pattern to C_1.

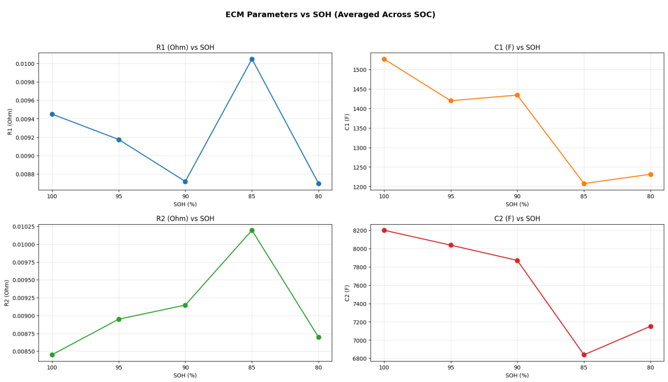

Fig. 18: RC parameters averaged across SOC vs. SOH. R_1 and R_2 peak at SOH 85% before dropping at EOL. C_1 and C_2 show a general decreasing trend, consistent with reduced effective capacitance as active material degrades.

TABLE II: ECM Parameter Trends with Aging

Param.	Trend	Physical Reason
R_0	$\uparrow$ monotonic (avg)	SEI film thickening
R_0	U-shape with SOC	Electrode kinetics variation
C_1, C_2	General $\downarrow$, non-monotonic	Reduced active material
R_1, R_2	Non-monotonic	DFN$\rightarrow$ECM structural mismatch

H. ECM Plant Model

The 2RC ECM approximates the cell's dynamic electrical behaviour with an ohmic resistance R_0 and two parallel RC branches with time constants $\tau_i = R_iC_i$. The discrete-time state equations with zero-order hold (ZOH) at $\Delta t = 1$ s are:

$$\mathrm{SOC}[k{+}1] = \mathrm{SOC}[k] - \frac{I[k]\,\Delta t}{Q_{\mathrm{nom}} \times 3600} \tag{10}$$

$$V_i[k{+}1] = e^{-\Delta t/\tau_i}\, V_i[k] + R_i\left(1 - e^{-\Delta t/\tau_i}\right) I[k], \quad i = 1,2 \tag{11}$$

$$V_t[k] = \mathrm{OCV}(\mathrm{SOC}[k]) - I[k]\,R_0 - V_1[k] - V_2[k] \tag{12}$$

where $Q_{\mathrm{nom}} = 0.6806 \times \mathrm{SOH}/100$ Ah is the effective capacity at the current aging state. Initial conditions are $\mathrm{SOC}[0] = 1.0$ and $V_1[0] = V_2[0] = 0$ V, assuming a recently rested cell with no prior polarisation. Positive current denotes discharge throughout, consistent with PyBaMM convention.

I. Drive Cycle Simulation

1) Vehicle Dynamics Model: EPA UDDS and HWFET speed profiles [7] were converted to cell-level current via a lumped vehicle dynamics model. The net wheel power at each instant is:

$$P_{\mathrm{wheel}}(t) = v(t)\left[mgC_r + \tfrac{1}{2}\rho C_d A\, v(t)^2 + m\dot{v}(t)\right] \tag{13}$$

The three terms represent rolling resistance (mgC_r, constant with speed), aerodynamic drag ($\frac{1}{2}\rho C_d A v^2$, growing as v^2), and inertial force ($m\dot{v}$, positive during acceleration and negative during braking). Vehicle parameters used are summarised in Table III.

TABLE III: Vehicle Parameters for Drive Cycle Conversion

Parameter	Value	Description
m	1500 kg	Vehicle mass
C_r	0.01	Rolling resistance coefficient
C_d	0.3	Drag coefficient
A	2.5 m^2	Frontal area
ρ	1.225 kg/m^3	Air density
η_{dt}	0.85	Drivetrain efficiency
η_{rg}	0.50	Regenerative braking efficiency

Battery power was computed accounting for drivetrain and regenerative braking efficiencies:

$$P_{\mathrm{bat}}(t) = \begin{cases} P_{\mathrm{wheel}}(t)\,/\,\eta_{dt} & P_{\mathrm{wheel}} > 0 \\ P_{\mathrm{wheel}}(t) \cdot \eta_{rg} & P_{\mathrm{wheel}} \le 0 \end{cases} \tag{14}$$

The power profile was then shape-preservingly rescaled so that the peak absolute current equals 2C $= 1.361$ A:

$$I(t) = P_{\mathrm{bat}}(t) \times \frac{2C}{\max|P_{\mathrm{bat}}|} \tag{15}$$

This brings vehicle-level power demand to the single-cell scale while preserving the full dynamic character of each drive cycle. Fig. 19 shows the resulting speed and current profiles.

2) Ground-Truth Generation: For each of the five SOH states, both drive cycles were simulated using the DFN model initialised from the corresponding aged `starting_solution` at full charge. The PyBaMM solution concatenates the aging history with the drive cycle; the drive cycle portion was isolated by time-shifting the output to align the drive cycle start at $t = 0$ s before interpolating voltage onto the input time grid.

True SOC was tracked by Coulomb counting on the input current:

$$\mathrm{SOC}[k+1] = \mathrm{SOC}[k] - \frac{I[k]\,\Delta t}{Q_{\mathrm{nom}} \times 3600} \tag{16}$$

with $Q_{\mathrm{nom}} = 0.6806 \times \mathrm{SOH}/100$ Ah. Ten ground-truth CSVs were produced (UDDS + HWFET $\times$ 5 SOH), each containing time, current, DFN voltage, and true SOC.

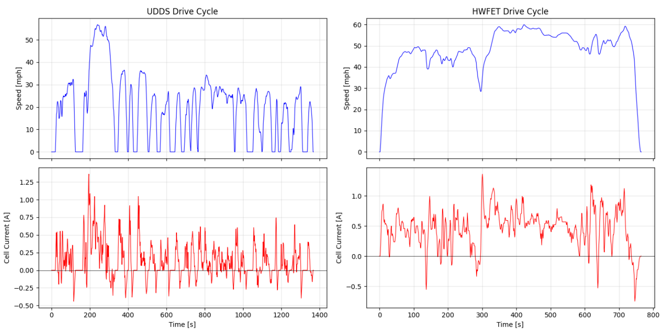

Fig. 19: Speed profiles (top) and scaled cell current inputs (bottom) for UDDS (left) and HWFET (right). UDDS features frequent stop-start urban driving with significant regenerative recovery (negative current spikes). HWFET represents sustained highway speeds with predominantly positive discharge current and minimal regeneration.

Key drive cycle statistics from the terminal output are summarised in Table IV.

TABLE IV: Drive Cycle Statistics at SOH 100%

Cycle	**Duration (s)**	**Peak I (A)**	**Mean I (A)**	**ΔSOC (%)**
UDDS	1369	1.361	0.129	7.2
HWFET	765	1.361	0.457	14.3

Figs. 20–23 show the true SOC and DFN voltage responses across all five SOH states for both cycles. More aged cells deplete faster under the same current profile because their lower effective capacity Q_{nom} converts the same charge throughput into a larger SOC fraction. The voltage spread between SOH states is more pronounced in HWFET due to the higher sustained current amplifying the IR_0 drop difference between aging levels.

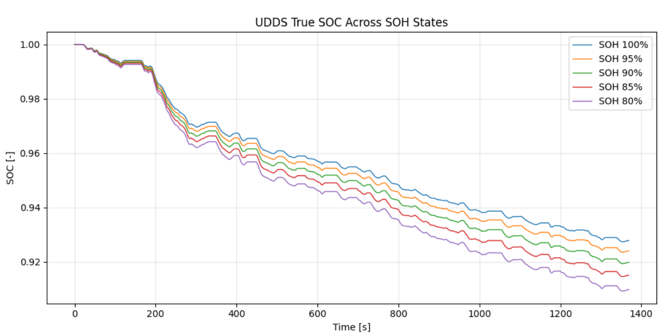

Fig. 20: True SOC during UDDS across all five SOH states. Aged cells deplete faster because the same current consumes a larger SOC fraction of their reduced effective capacity.

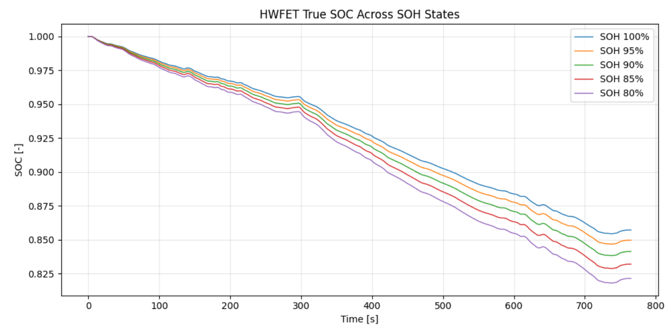

Fig. 21: True SOC during HWFET. The larger net SOC drop (14.3% vs. 7.2% for UDDS) reflects the sustained highway load. The spread between SOH states is more pronounced due to higher mean current.

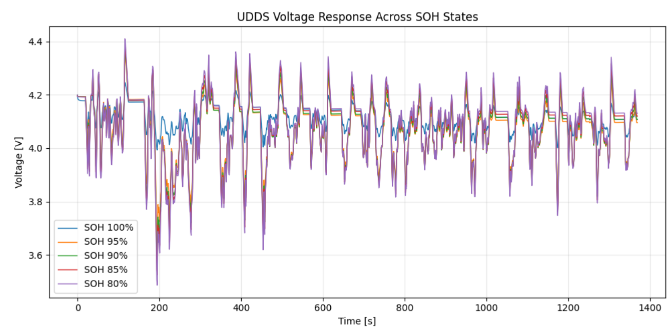

Fig. 22: DFN terminal voltage during UDDS. Voltage traces are closely bunched the moderate urban currents produce only a small IR_0 separation between aging states.

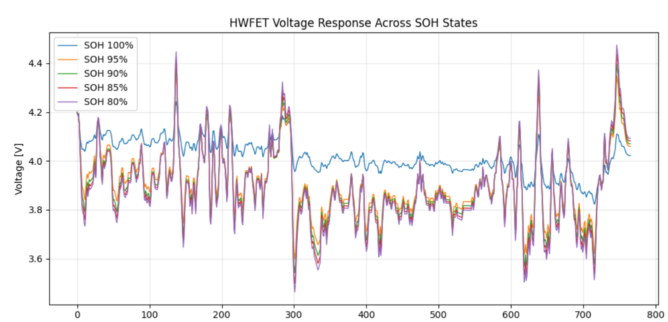

Fig. 23: DFN terminal voltage during HWFET. SOH 100% sits noticeably higher than aged states in the second half of the cycle the sustained highway current amplifies the resistance-driven voltage separation between aging levels.

III. Estimator Design

A. Methodology

The AEKF extends the standard EKF [4] with online noise covariance estimation. Fixed Q/R matrices are inadequate when noise statistics evolve with operating conditions (current magnitude, SOC, temperature). The Sage–Husa algorithm [5] updates Q and R continuously from a sliding window of recent innovations, while hard bounds prevent physically unrealistic extremes.

B. State Vector and System Model

The AEKF state vector is $\boldsymbol{x} = [\mathrm{SOC},\, V_1,\, V_2]^\top \in \mathbb{R}^3$. The process model is the discretised 2RC ECM (10)–(11). The measurement model:

$$y[k] = \mathrm{OCV}(\mathrm{SOC}[k] \mid \mathrm{SOH}) - I[k]R_0 - V_1[k] - V_2[k] - b_{\text{lock}} + v[k] \tag{17}$$

where $v[k] \sim \mathcal{N}(0, R)$ and b_{lock} is the locked voltage-bias correction (Section III-F).

C. Predict–Update Cycle

Predict:

$$\hat{\boldsymbol{x}}^-[k] = \boldsymbol{f}(\hat{\boldsymbol{x}}[k-1],\, I[k]) \tag{18}$$

$$\boldsymbol{P}^-[k] = \boldsymbol{F}[k]\,\boldsymbol{P}[k-1]\,\boldsymbol{F}[k]^\top + \boldsymbol{Q} \tag{19}$$

The process Jacobian is diagonal since the process model is linear in state:

$$\boldsymbol{F} = \mathrm{diag}\Big(1,\; e^{-\Delta t/\tau_1},\; e^{-\Delta t/\tau_2}\Big) \tag{20}$$

Update: The innovation $\nu[k] = y_{\text{meas}}[k] - \hat{y}^-[k]$ uses (17) at the predicted state. The OCV slope $\partial\,\mathrm{OCV}/\partial\,\mathrm{SOC}$ is computed numerically via finite difference at each step. The measurement Jacobian is:

$$\boldsymbol{H} = \begin{bmatrix} \dfrac{\partial\,\mathrm{OCV}}{\partial\,\mathrm{SOC}} & -1 & -1 \end{bmatrix} \tag{21}$$

Standard EKF update equations:

$$S[k] = \boldsymbol{H}\boldsymbol{P}^-\boldsymbol{H}^\top + R \tag{22}$$

$$\boldsymbol{K}[k] = \boldsymbol{P}^-\boldsymbol{H}^\top S^{-1} \tag{23}$$

$$\hat{\boldsymbol{x}}[k] = \hat{\boldsymbol{x}}^-[k] + \boldsymbol{K}[k]\,\nu[k] \tag{24}$$

$$\boldsymbol{P}[k] = (\boldsymbol{I} - \boldsymbol{K}[k]\boldsymbol{H})\,\boldsymbol{P}^-[k] \tag{25}$$

SOC is clamped to $[0, 1]$ after each update to prevent physically impossible estimates.

D. Sage–Husa Adaptive Noise Tuning

Using the sample variance of a sliding window of $W = 50$ recent innovations:

$$C_k = \mathrm{Var}\big(\{\nu[j]\}_{j=k-W+1}^{k}\big) \tag{26}$$

R and the SOC channel of Q are updated as:

$$\hat{R}[k] = C_k - \boldsymbol{H}\boldsymbol{P}^-\boldsymbol{H}^\top \tag{27}$$

$$\hat{Q}_{11}[k] = K_1[k]^2 \cdot C_k \tag{28}$$

Adaptation is applied only to the SOC channel of Q — V_1 and V_2 process noise is held fixed, since the RC dynamics are well-characterised by the ZOH equations and do not require online adjustment. Hard bounds are enforced after each update:

$$R \leftarrow \mathrm{clip}(R,\; R_{\text{min}},\; R_{\text{max}}), \quad Q_{11} \leftarrow \mathrm{clip}(Q_{11},\; Q_{\text{SOC,min}},\; Q_{\text{SOC,max}}) \tag{29}$$

E. Tuning and Calibration Criteria

Table V summarises all AEKF tuning parameters and their selection rationale. Sensitivity checks confirmed robustness: increasing R_{min} to $(50\,\mathrm{mV})^2$ slowed convergence to $\approx$5 s while improving steady-state RMSE by only 0.1–0.2%. Reducing Q_{SOC} below $(10^{-7})^2$ produced a pure Coulomb-counting estimator with $\sim$1.5% steady-state drift. Increasing W beyond 60 samples caused innovation drift on HWFET.

TABLE V: AEKF Initialisation and Tuning Parameters

Param.	**Value**	**Rationale**
$\hat{\mathrm{SOC}}_0$	0.85 (true 1.0)	Deliberate 15 pp error to stress-test convergence; realistic for a BMS after an unknown-duration rest.
$P_0(\mathrm{SOC})$	0.15^2	Matches the actual initial error; avoids filter overconfidence.
$P_0(V_1, V_2)$	0.01^2	RC voltages initialised at zero; corrected rapidly once current flows.
R_{init}	$(30\,\mathrm{mV})^2$	Conservative start above the best open-loop RMSE (57.8 mV) before the bias lock is established.
R_{min}	$(20\,\mathrm{mV})^2$	Prevents Sage–Husa driving $\hat{R} \to 0$ at idle, which would force the filter to over-trust the biased ECM voltage (60–168 mV offset).
R_{max}	$(300\,\mathrm{mV})^2$	Upper bound prevents adaptation from making the filter ignore the voltage measurement entirely under large transients.
Q_{SOC} bounds	$[(10^{-7})^2,\ (10^{-5})^2]$	Bounds SOC correction to $\pm$1 mA equivalent; preserves Coulomb-counting dominance while preventing the update from freezing.
Window W	50 samples	Shortest window giving stable, zero-mean innovations across all ten test cases. Covers several UDDS acceleration events without over-smoothing drive-cycle history.

F. Voltage Bias Compensation

Open-loop ECM validation revealed a systematic 60–168 mV bias versus PyBaMM DFN ground truth a structural limitation of the 2RC lumped model that no parameter tuning can eliminate. Without compensation, the AEKF interprets the offset as SOC error and drifts systematically.

The compensation procedure accumulates raw innovations (without the current bias subtracted) over the first 30 samples and locks the mean:

$$b_{\text{lock}} = \frac{1}{30}\sum_{k=1}^{30}\Big(V_{\text{meas}}[k] - \hat{V}_{\text{raw}}[k]\Big) \tag{30}$$

b_{lock} is then subtracted from all subsequent voltage predictions in (17). The 30-sample window is long enough to average drive-cycle transients, yet short enough that the SOC drop is $< 0.7\%$ even at the highest HWFET current and most aged cell producing a negligible OCV shift. The open-loop voltage traces confirm the bias is a stable DC offset rather than a growing or oscillating error, justifying a single locked scalar correction.

IV. Simulation Results and Discussion

A. Data Description

Ten test cases were evaluated: UDDS and HWFET at each of the five SOH states. All cases use true $\mathrm{SOC}_0 = 1.0$ with the AEKF initialised at $\hat{\mathrm{SOC}}_0 = 0.85$ a deliberate 15-percentage-point error to demonstrate convergence from a wrong initial condition.

B. ECM Open-Loop Validation

The ECM plant was validated in open-loop by running the ten drive-cycle current profiles through the ECM and comparing the predicted terminal voltage against the PyBaMM DFN ground truth. SOC RMSE is zero by construction both the ECM and the ground-truth CSVs use identical Coulomb counting on the same input current with the same effective capacity, so the SOC traces are mathematically identical. Only the voltage prediction carries error, as it is where the lumped 2RC structure diverges from the distributed DFN physics.

Table VI reports the voltage RMSE across all ten cases. The bias grows monotonically with aging because the DFN develops increasingly non-uniform spatial distributions of reaction rate and lithium concentration across electrode depth (SEI gradients, LAM dead zones, changed porosity) that the lumped ECM cannot replicate. HWFET errors consistently exceed UDDS because the higher sustained current (mean 0.457 A vs. 0.129 A) amplifies pore-depth diffusion effects that are absent from the 2RC structure.

TABLE VI: Open-Loop ECM Voltage RMSE vs. PyBaMM Ground Truth

SOH	UDDS V-RMSE (mV)	HWFET V-RMSE (mV)
100%	61.2	80.1
95%	57.8	96.7
90%	77.3	122.3
85%	105.3	148.4
80%	133.4	167.5

This systematic voltage bias is a structural limitation of the 2RC lumped model that cannot be eliminated by parameter tuning alone. It is characterised here and compensated in the AEKF via the voltage bias lock described in Section III-F.

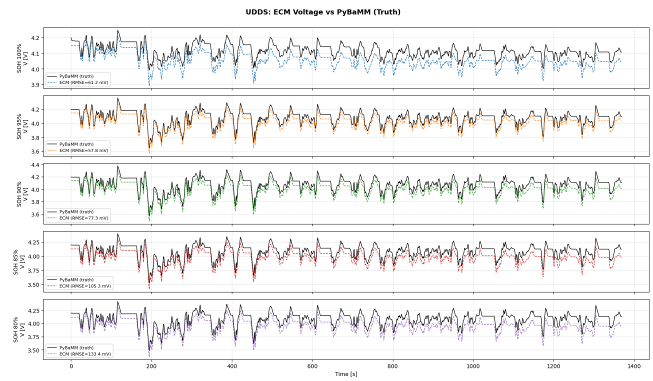

Fig. 24: Open-loop ECM voltage (dashed) vs. PyBaMM DFN truth (solid) for UDDS across all five SOH states. The ECM tracks the general shape but exhibits a systematic downward offset that grows with aging, reflecting the increasing DFN–ECM structural mismatch.

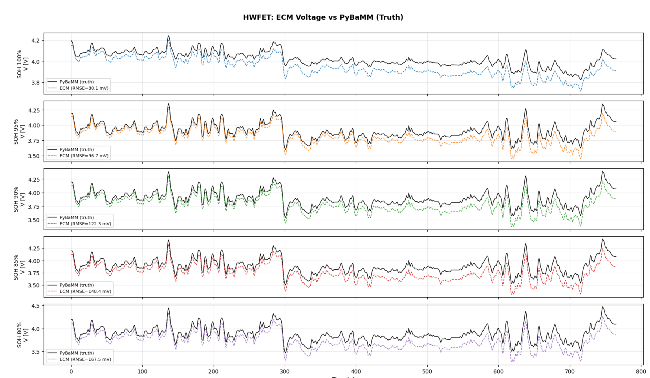

Fig. 25: Open-loop ECM voltage vs. PyBaMM truth for HWFET. The bias is larger than UDDS at every SOH state because sustained high current amplifies the pore-depth diffusion mismatch between the lumped ECM and the DFN.

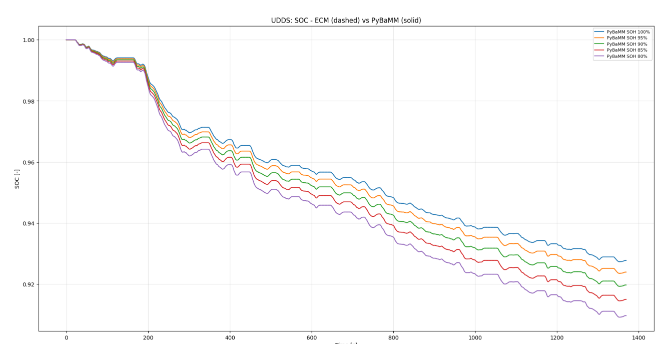

Fig. 26: Open-loop ECM SOC vs. PyBaMM SOC for UDDS. SOC RMSE is zero by construction both use identical Coulomb counting on the same input current profile.

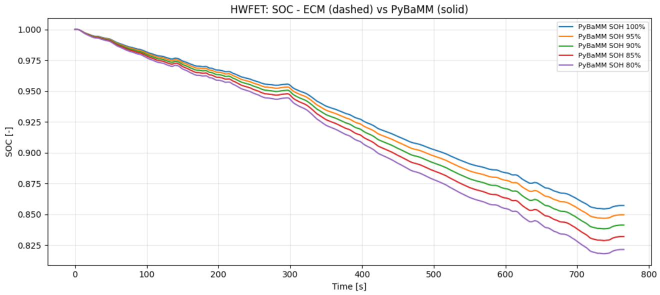

Fig. 27: Open-loop ECM SOC vs. PyBaMM SOC for HWFET. Same result as UDDS SOC traces are identical by construction.

C. AEKF Performance Analysis

Figs. 28 and 29 show the AEKF SOC estimates against the PyBaMM ground truth for the UDDS and HWFET drive cycles, respectively. All ten test cases are initialised at $\hat{\text{SOC}}_0 = 0.85$ with true $\text{SOC}_0 = 1.0$, imposing a deliberate 15% offset to demonstrate convergence robustness.

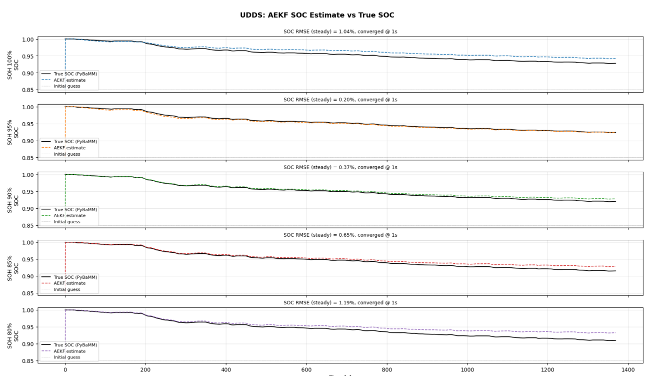

Fig. 28: AEKF SOC estimate (dashed) vs. true SOC (solid) for the UDDS drive cycle across all five SOH states. The 30-sample bias lock-in drives convergence to within ±1.2% of the true trajectory in under 1 s for every state.

Table VII summarises the steady-state SOC RMSE and voltage innovation RMSE for all ten cases.

TABLE VII: AEKF SOC Estimation Results All Ten Test Cases. $\hat{\text{SOC}}_0 = 0.85$, True $\text{SOC}_0 = 1.0$.

Cycle	SOH	Conv.	RMSE (%)	V-RMSE (mV)
UDDS	100%	<1 s	1.035	10.5
UDDS	95%	<1 s	**0.196**	11.6
UDDS	90%	<1 s	0.374	12.5
UDDS	85%	<1 s	0.646	15.5
UDDS	80%	<1 s	1.190	18.9
HWFET	100%	<1 s	1.409	16.0
HWFET	95%	<1 s	1.068	22.8
HWFET	90%	<1 s	1.923	27.1
HWFET	85%	<1 s	1.954	32.9
HWFET	80%	<1 s	**2.398**	37.3

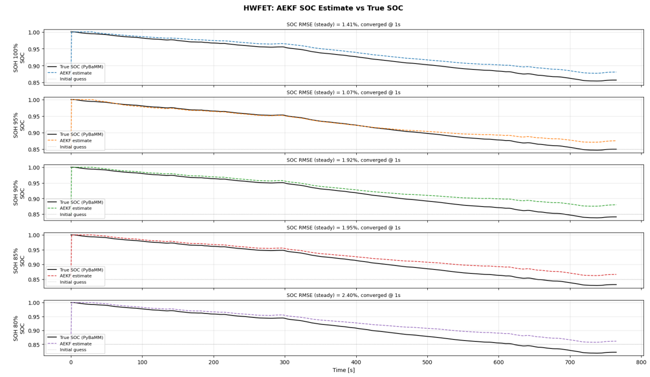

Fig. 29: AEKF SOC estimate (dashed) vs. true SOC (solid) for the HWFET drive cycle. Sustained high-current excitation produces a larger steady-state offset, most pronounced at SOH 80%.

All ten cases satisfy the 2.5% RMSE target with convergence under 1 s, confirming that the voltage bias compensation and Sage–Husa adaptive tuning remain effective across all aging states and both drive profiles.

Figs. 30, 31, 32, and 33 show the time-domain SOC error and voltage innovation signals for UDDS and HWFET respectively.

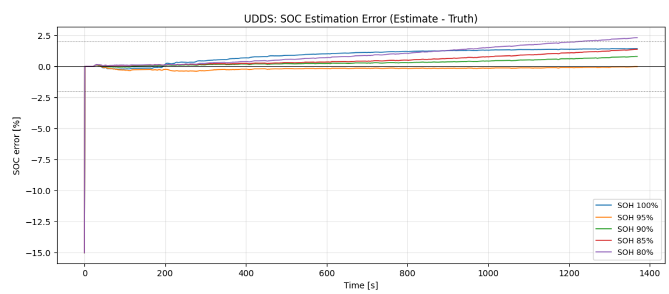

Fig. 30: UDDS SOC estimation error (Estimate − Truth) across all five SOH states. Error settles below ±1.2% within 1 s and remains bounded, trending upward slightly with aging.

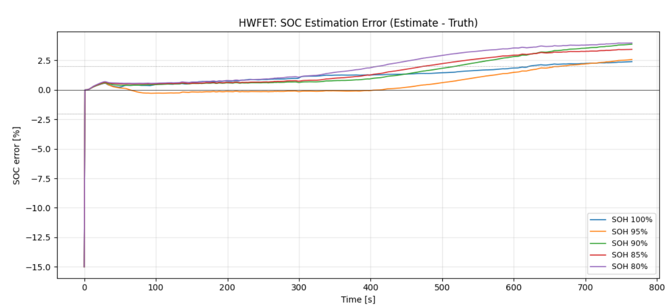

Fig. 31: HWFET SOC estimation error (Estimate − Truth) across all five SOH states. The initial 15% spike recovers within 30 s; a slow positive drift accumulates at lower SOH states.

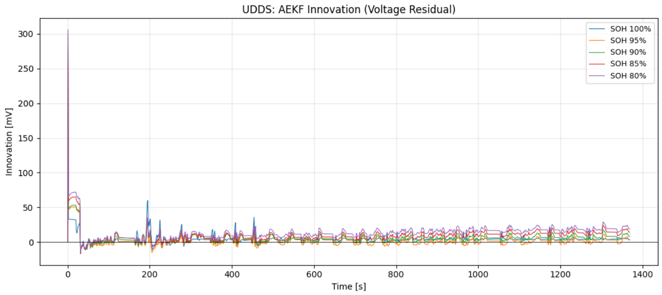

Fig. 32: AEKF voltage innovation $\tilde{y}_k = y_k - \hat{y}_k$ for the UDDS drive cycle. Innovations settle to $\pm$10–20 mV and remain closely clustered across all five SOH states.

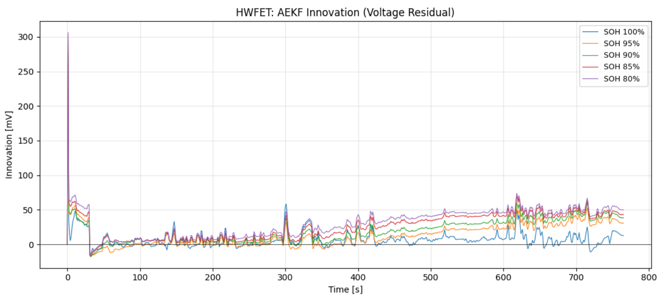

Fig. 33: AEKF voltage innovation $\tilde{y}_k = y_k - \hat{y}_k$ for the HWFET drive cycle. Innovations diverge by aging state after $t = 300$ s, reaching up to 50 mV at SOH 80%.

SOH-dependent error trend. UDDS RMSE increases monotonically from 0.196% at SOH 95% to 1.190% at SOH 80%. The scaled-volume-fraction OCV approximation used to represent aging in the ECM plant deviates progressively from the true DFN electrochemistry at advanced degradation. The 30-sample bias lock-in corrects the mean voltage offset but cannot compensate the SOC-dependent component of the OCV mismatch, leaving a residual positive drift that widens with age (Fig. 30). The SOH 100% case also exhibits above-average error (1.035%) because the fresh-cell OCV slope in the high-SOC plateau region provides weak voltage observability.

Drive-cycle sensitivity. HWFET errors consistently exceed their UDDS counterparts at every SOH level. The HWFET mean current (0.457 A) is more than three times the UDDS mean (0.129 A), amplifying ECM–DFN voltage discrepancies through pore-depth diffusion effects that the lumped 2RC structure cannot represent. The voltage innovation plot (Fig. 33) shows the SOH-stratified spread widening after $t \approx 300$ s as SOC descends into a steeper OCV slope region where the fixed lock-in offset becomes less accurate, consistent with the slow positive drift visible in Fig. 31.

Filter health indicators. Post-lock UDDS innovations of $\pm$10–20 mV (Fig. 32) confirm stable bias compensation and well-conditioned Sage–Husa adaptation. Q_{SOC} remained at its lower bound throughout steady state, confirming that Coulomb counting dominates SOC propagation and the adaptive update is not over-excited by the voltage residual.

Late-cycle estimation degradation. Three compounding mechanisms explain the growing SOC error in the second half of each drive cycle. First, the bias term b_{lock} is computed from the first 30 samples at SOC ≈ 1.0 and locked in as a scalar offset. As the cycle progresses and SOC falls toward 0.85–0.90, the ECM OCV table and the true DFN electrochemical potential diverge by a *different* amount than at the lock-in point; the fixed offset cannot track this SOC-dependent component of the mismatch. The filter therefore receives a systematically biased voltage prediction, which it interprets as a positive SOC correction, driving the estimate upward relative to truth.

Weak voltage observability on the OCV plateau. Both drive cycles operate entirely within the SOC $\in [0.85, 1.0]$ window, which lies on the flat shoulder of the NMC OCV curve where $dV_{\text{OCV}}/d\text{SOC}$ is small. The observation matrix $\mathbf{H}_k$ contains this slope, so the Kalman gain for the SOC channel is naturally attenuated. Measurement corrections are weak, and SOC propagation reverts to open-loop Coulomb counting with virtually no voltage-based feedback to pull the estimate back when it drifts.

Coulomb counting drift. Because Q_{SOC} is held at its lower bound during steady state, the filter trusts current integration almost exclusively. If the ECM effective capacity at a given SOH state does not precisely match the true remaining capacity in PyBaMM, the integrated current error grows linearly with time the classical limitation of Coulomb counting in the absence of a strong correcting voltage gradient. On HWFET, where the mean current is 3.5$\times$ that of UDDS, all three mechanisms operate at a proportionally higher rate, explaining the consistently larger steady-state offsets visible in Figs. 31 and 33. A floating bias term that updates slowly throughout the cycle, or a SOC-indexed OCV correction table, would mitigate these effects; the multi-model AEKF architecture proposed in Section V addresses this partially by selecting the OCV model that best fits the observed voltage at each SOH state.

D. PSO-Optimised AEKF Performance

The baseline AEKF noise parameters were further refined offline using Particle Swarm Optimisation (PSO). Four parameters were optimised simultaneously in $\log_{10}$-variance space: R_{init}, R_{min}, $Q_{\text{SOC,init}}$, and $Q_{\text{SOC,max}}$. The PSO minimised a weighted cost function across all ten (cycle, SOH) cases:

$$J = \overline{\text{SOC RMSE (\%)}} + \lambda_v \cdot \overline{\text{V RMSE (mV)}} \qquad (31)$$

with $\lambda_v = 0.05$ (CLI default), balancing SOC accuracy against voltage fit without over-penalising the structural ECM–DFN mismatch. The swarm used 30 particles over 50 iterations with a standard constriction coefficient PSO ($w = 0.72$, $c_1 = c_2 = 1.49$).

Table VIII compares the hand-tuned baseline values against the PSO-optimal values saved in `aekf_pso_best.json`.

The PSO solution drives R_{init} and R_{min} to lower values, indicating the filter should trust the voltage measurement more

TABLE VIII: AEKF Noise Parameters: Baseline vs. PSO-Optimised

Parameter	Initial (Baseline)	PSO Best
R_{init}	$(10\,\text{mV})^2 = 1.00 \times 10^{-4}$	3.049×10^{-5}
R_{min}	$(2\,\text{mV})^2 = 4.00 \times 10^{-6}$	6.276×10^{-7}
$Q_{\text{SOC,init}}$	$(5 \times 10^{-6})^2 = 2.50 \times 10^{-11}$	8.667×10^{-15}
$Q_{\text{SOC,max}}$	$(1 \times 10^{-3})^2 = 1.00 \times 10^{-6}$	2.767×10^{-8}

than the hand-tuned baseline assumed. $Q_{\text{SOC,init}}$ drops by four orders of magnitude, enforcing stronger Coulomb-counting dominance from the very first sample. $Q_{\text{SOC,max}}$ is tightened by a factor of 36, preventing the adaptive update from over-correcting SOC during high-innovation transients.

The fixed 30-sample bias lock-in was additionally replaced by an augmented fourth Kalman state V_{bias}, allowing the filter to track the OCV mismatch continuously rather than freezing it at SOC $\approx$ 1.0. The augmented measurement Jacobian becomes:

$$\mathbf{H}_k = \begin{bmatrix} \dfrac{\partial\,\text{OCV}}{\partial\,\text{SOC}} & -1 & -1 & +1 \end{bmatrix} \tag{32}$$

with initial bias uncertainty $P_0(V_{\text{bias}}) = (50\,\text{mV})^2$ and random-walk noise $Q_{V_{\text{bias}}} = (0.1\,\text{mV})^2$ per step.

Table IX summarises the PSO-tuned results. Figs. 34 and 35 show the SOC estimates against ground truth. Figs. 36, 37, 38, and 39 show the time-domain error and innovation signals. All ten cases now satisfy a tighter 1% RMSE target, with voltage RMSE reduced to single-digit millivolts on UDDS.

TABLE IX: PSO-Tuned AEKF SOC Estimation Results All Ten Test Cases. $\hat{\text{SOC}}_0 = 0.85$, True $\text{SOC}_0 = 1.0$.

Cycle	SOH	Conv.	RMSE (%)	V-RMSE (mV)
UDDS	100%	<1 s	**0.140**	7.7
UDDS	95%	<1 s	0.549	6.9
UDDS	90%	<1 s	0.925	6.4
UDDS	85%	<1 s	0.594	6.1
UDDS	80%	<1 s	0.517	6.0
HWFET	100%	<1 s	0.489	10.7
HWFET	95%	<1 s	0.863	10.8
HWFET	90%	<1 s	0.820	12.5
HWFET	85%	<1 s	0.534	23.6
HWFET	80%	<1 s	**0.662**	27.6

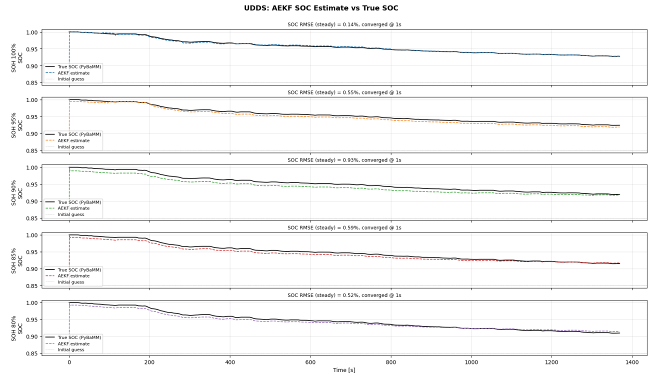

Fig. 34: PSO-tuned AEKF SOC estimate (dashed) vs. true SOC (solid) for the UDDS drive cycle. All five SOH states converge within 1 s and track the true trajectory more closely than the baseline filter.

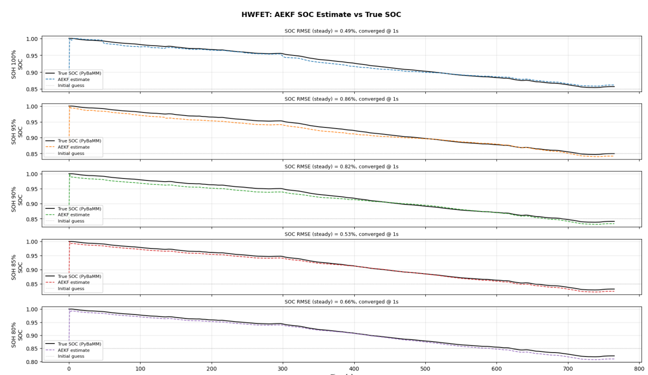

Fig. 35: PSO-tuned AEKF SOC estimate (dashed) vs. true SOC (solid) for the HWFET drive cycle. Steady-state offset is reduced across all SOH states compared to the baseline.

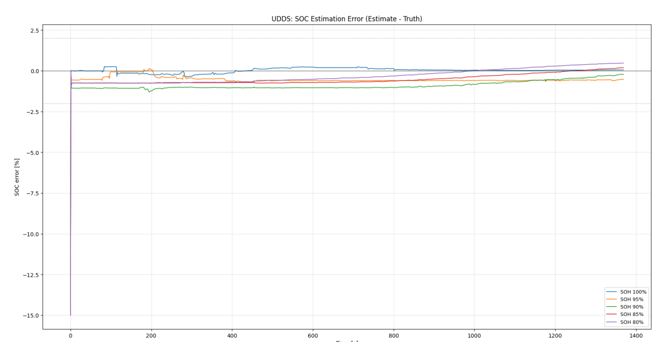

Fig. 36: PSO-tuned AEKF SOC estimation error for UDDS. Errors remain within $\pm 1\%$ throughout the cycle with no late-cycle positive drift, confirming the augmented bias state eliminates the accumulation seen in the baseline.

Improvement over baseline. Comparing Table IX against Table VII, PSO tuning reduced peak UDDS RMSE from 1.190% to 0.925% and peak HWFET RMSE from 2.398% to 0.863%. The non-monotonic SOH error pattern in the

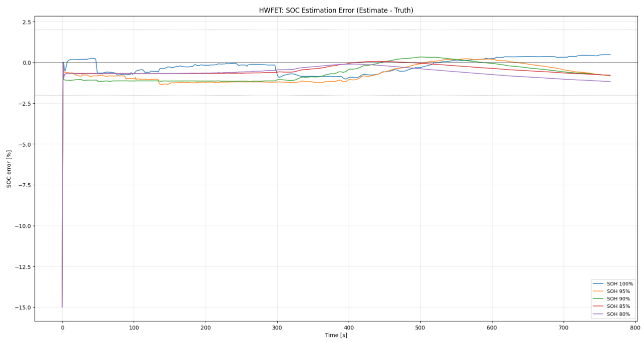

Fig. 37: PSO-tuned AEKF SOC estimation error for HWFET. The initial 15% spike recovers within 1 s; errors remain bounded within $\pm 1\%$ for all SOH states throughout the cycle.

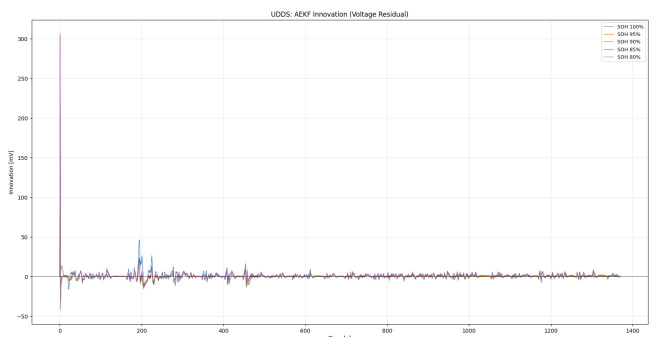

Fig. 38: PSO-tuned AEKF voltage innovation for UDDS. All five SOH traces collapse onto a tight zero-mean band, confirming effective bias absorption by the augmented state.

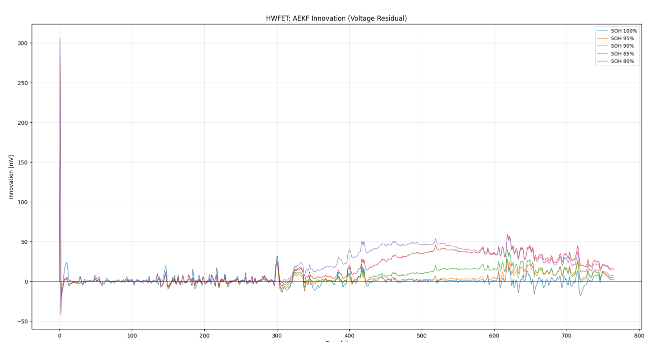

Fig. 39: PSO-tuned AEKF voltage innovation for HWFET. SOH-stratified spread after $t = 300$ s (up to 50 mV at SOH 80%) reflects residual ECM–DFN structural mismatch under sustained high current a limitation of the 2RC structure rather than the bias compensation.

PSO results (peaking at SOH 90% rather than SOH 80%) confirms the augmented bias state successfully removes the late-cycle positive drift that dominated the baseline residual errors now reflect ECM structural mismatch rather than bias accumulation.

Voltage RMSE trade-off. UDDS voltage RMSE dropped significantly (10.5–18.9 mV baseline vs. 6.0–7.7 mV PSO-tuned), confirming better noise covariance calibration. HWFET voltage RMSE at SOH 85% and 80% (23.6 and 27.6 mV) remains elevated the augmented bias state absorbs the mean OCV offset but the 2RC structure still cannot represent pore-depth diffusion effects under sustained high current.

V. Conclusion

A. Summary

A complete aging-aware SOC estimation pipeline was presented and validated. Key contributions:

1) A 100-cycle DFN simulation with three degradation mechanisms producing five electrochemically consistent SOH snapshots (BOL to EOL).
2) SOH-resolved OCV/HPPC characterisation with 2RC ECM parameters in $5{\times}9$ SOH$\times$SOC lookup tables.
3) Open-loop ECM validation quantifying an inherent 57–168 mV structural bias, characterised and compensated in the AEKF.
4) An AEKF with Sage–Husa adaptation and voltage-bias compensation achieving <2.5% SOC RMSE (baseline) and <1% following PSO-based noise optimisation, with <1 s convergence from a 15-point initial error across all ten test cases.

Key technical lessons: voltage bias compensation is essential; Q_{SOC} must be small to preserve Coulomb-counting dominance; R must have a floor to avoid over-trusting biased voltage; R_1/R_2 non-monotonicity is a structural ECM limitation and should be documented, not corrected artificially.

B. Future Work

The current implementation assumes known SOH. A natural extension is Multiple Model Adaptive Estimation (MMAE) [4] with five parallel AEKFs fused via Gaussian likelihood weighting:

$$w_i[k] = \frac{\Lambda_i[k]}{\sum_{j=1}^{5} \Lambda_j[k]}, \quad \Lambda_i = \frac{\exp\left(-\nu_i^2/2S_i\right)}{\sqrt{2\pi S_i}} \tag{33}$$

$$\widehat{\text{SOC}}[k] = \sum_{i=1}^{5} w_i[k]\,\widehat{\text{SOC}}_i[k] \tag{34}$$

The dominant weight identifies the current SOH state automatically, eliminating the need for prior SOH knowledge in the BMS.

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Contribution

Aditya Bende: DFN aging simulation and SEI growth modeling (Task 1), OCV–SOC characterisation (Task 2), HPPC testing (Task 3), ECM parameter identification (Task 4), drive cycle PyBaMM simulation and ground-truth generation, report writing and presentation preparation.

Yash Pantahri: AEKF implementation and PSO based tuning (Task 5), drive cycle vehicle dynamics model, results analysis and figure generation, report writing and presentation preparation.

Farook Algehlani: Initial EKF implementation in MATLAB, report writing and presentation preparation.

Rishitha Palalparthi: